Non-Negative Tensor Decomposition using Monotonic FISTA and Dampened Residual Tracking

Thejus S, Madhu S. Nair

***Abstract*—Standard Non-Negative Tensor Decomposition (NTD-tSVD) is fundamentally bottlenecked by *Projection Amnesia*—a phenomenon where greedy rank-truncation repeatedly discards compression residuals, permanently capping reconstruction fidelity. To overcome this limitation, we introduce Hybrid FISTA-RT, a novel architecture integrating Dampened Residual Tracking to recycle discarded errors, Monotonic FISTA Acceleration to expedite convergence, and strict Output Boundary Enforcement. Evaluated across continuous rank spectrums, the proposed framework demonstrates robust superiority. Specifically at rank $R = 30$, our method achieves Peak Signal-to-Noise Ratios (PSNR) of 31.47 dB on the COIL-20 visual dataset and an unprecedented 111.18 dB on the Enron sparse temporal graph, dominating state-of-the-art alternating baselines across all domains.**

***Index Terms*—Tensor Decomposition, Fast Iterative Shrinkage-Thresholding Algorithm (FISTA), Dampened Residual Tracking, Projection Amnesia**

I. INTRODUCTION

MODERN computer vision, autonomous navigation, and medical imaging rely heavily on the rapid processing of high-dimensional visual data. Because real-world sensor data is inherently noisy and sparse, efficient low-rank reconstruction is critical for preserving structural fidelity in downstream machine learning tasks [1], [6], [14].

Literature Review. Traditional matrix-based recovery methods (e.g., PCA, SVD) necessitate flattening multi-dimensional data, destroying vital spatial and temporal correlations [13]. To preserve this multi-way structure, tensor decompositions such as CP and Tucker [1], [11] became standard. More recently, the tensor singular value decomposition (t-SVD) framework [8] introduced tubal-rank minimization, offering superior data recovery [9]. Building on this, the Non-Negative Tensor Decomposition (NTD-tSVD) architecture [2] was proposed to enforce physical viability (e.g., zero-background constraints for sparse images). While optimization frameworks like FISTA [3], [10], [12] successfully accelerate convex matrix problems, applying them to the non-convex rank-truncation of tensors remains highly unstable.

The Problem: Projection Amnesia. While the standard NTD-tSVD baseline effectively isolates signals—achieving a benchmark PSNR of 26.30 dB on MNIST [4]—it is bottlenecked by a critical algorithmic flaw we term *Projection Amnesia*. At each alternating projection step, the algorithm employs greedy rank-truncation that blindly discards compression residuals. This persistent loss of historical error data stalls the optimization loop and permanently caps reconstruction fidelity, preventing reliable deployment in high-precision environments.

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Proposed Methodology & Contributions. To overcome this hard mathematical ceiling, we introduce **Hybrid FISTA-RT**, a novel tensor reconstruction architecture optimized for high-dimensional GPU scalability [5]. By actively recycling truncation errors and enforcing strict domain boundaries, our framework supersedes the standard Alternating Projections baseline. Our core contributions are:

- **Dampened Residual Tracking via Alpha Decay:** We introduce a fading-memory tensor $\mathcal{Q}$ that captures and recycles rank-truncation errors to cure projection amnesia. Governed by an exponential decay factor ($\alpha_k = \alpha_0 \cdot \gamma^k$), this mechanism heavily weights memory during initial iterations to structurally repair artifacts, then smoothly phases out to allow unhindered late-stage fine-tuning.
- **Monotonic FISTA Acceleration:** We equip an accelerated momentum framework with a strict monotonic safety brake. If reconstruction fitness drops, the algorithm instantly reverts to the last safe state, zeroes its momentum, and softens the memory bank, guaranteeing stable descent without the catastrophic momentum crashes common in non-convex tracking.
- **Empirical Dominance & Semantic Validation:** Evaluated across continuous compression ranks ($R = 1$ to 70), our architecture achieves a peak PSNR of 27.11 dB at $R = 60$ on MNIST (a 0.81 dB improvement over the state-of-the-art baseline). At $R = 30$, it achieves 31.47 dB on COIL-20 and an unprecedented 111.18 dB on the Enron sparse temporal graph, while downstream CNN classification tests confirm its superior preservation of semantic structural fidelity.

The remainder of this paper is organized as follows. Section II presents the Hybrid FISTA-RT model. Section III analyzes complexity and stability. Section IV reports empirical experiments. Section V concludes the paper.

Notation

Scalars are denoted by lowercase letters (e.g., t, α, f), vectors by bold lowercase (e.g., $\mathbf{v}$), and 3D tensors by calligraphic letters (e.g., $\mathcal{X}, \mathcal{M}, \mathcal{Q}$). For a tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$, its Frobenius norm is $\|\mathcal{X}\|_F$. Element-wise multiplication is denoted by $\odot$, and the tensor-tensor product (t-product) by $*$. The target truncation rank is R.

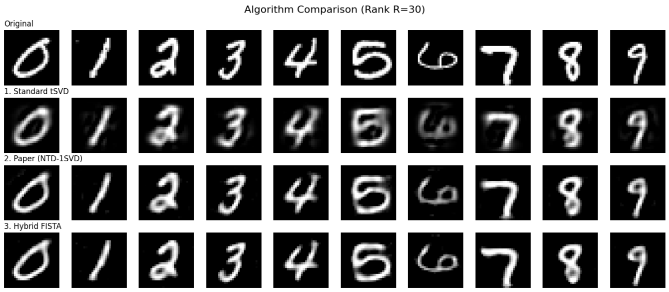

Fig. 1. Low-rank approximation for the MNIST dataset (784 × 5000 × 10) with rank $R = 30$. **Row 1:** Original dataset. **Row 2:** Standard tSVD. **Row 3:** NTD-tSVD (26.30 dB). **Row 4:** Hybrid FISTA (27.80 dB).

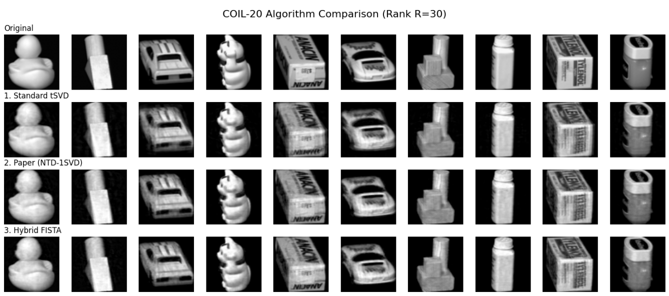

Fig. 2. Visual reconstruction quality comparison on the COIL-20 dataset at target rank $R = 30$. **Row 1:** Original uncompressed images. **Row 2:** Standard tSVD (28.873 dB). **Row 3:** NTD-1SVD baseline (30.653 dB). **Row 4:** Proposed Hybrid FISTA (31.474 dB).

During optimization at iteration k, $\mathcal{M}^{(k)}$ is the physically constrained tensor state, $\mathcal{Y}^{(k)}$ is the FISTA momentum-extrapolated search space, $\mathcal{Q}^{(k)}$ is the dampened residual tracking (memory) tensor, and $\mathcal{P}^{(k)}$ is the unbounded low-rank approximation outputted by the truncated t-SVD. Ω_1 and Ω_0 denote boolean index masks for the foreground and background constraints, respectively.

II. Proposed Methodology

A. Problem Formulation

Given a nonnegative sparse tensor $\mathcal{M} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$, we seek a low-rank latent tensor $\mathcal{X}$ such that

$$\mathcal{M} \approx \max(0, \mathcal{X}), \quad \mathrm{rank}_t(\mathcal{X}) \leq R, \tag{1}$$

which is the ReLU-NTD formulation of [2]. The reconstruction loss in the rectified domain is

$$\min_{\mathcal{X}} \ \tfrac{1}{2} \|\mathcal{M} - \max(0, \mathcal{X})\|_F^2 \quad \text{s.t.} \quad \mathrm{rank}_t(\mathcal{X}) \leq R. \tag{2}$$

The standard NTD-tSVD baseline solves (2) by alternating projection, but the greedy W-step discards the projection residual at every iteration, creating the Projection Amnesia bottleneck.

B. Architectural Pillars

1) Dampened Residual Tracking via Alpha Decay: To eliminate Projection Amnesia, a fading-memory tensor $\mathcal{Q} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$, initialized to zero, is introduced. At iteration k, let $\mathcal{P}^{(k)}$ denote the raw, unbounded t-rank-R approximation outputted by the SVD, and $\mathcal{M}^{(k)}$ the physically constrained projection. The tracking tensor captures the residual difference:

$$\mathcal{Q}^{(k+1)} = \alpha_k \Big(\mathcal{Q}^{(k)} + \mathcal{P}^{(k)} - \mathcal{M}^{(k)} \Big), \tag{3}$$

where α_k is governed by an exponential decay function:

$$\alpha_k = \alpha_0 \cdot \gamma^k. \tag{4}$$

Here, α_0 represents the initial memory weight and $\gamma \in (0, 1)$ is the decay rate. This ensures the algorithm heavily relies on historical tracking to structurally repair large artifacts during early iterations, but smoothly phases out the memory bank to allow unhindered, precision fine-tuning in the final stages. The stored residual $\mathcal{Q}^{(k+1)}$ is injected into the target variable of the subsequent loop, forcing the algorithm to actively self-correct.

2) Monotonic FISTA Acceleration: To accelerate global convergence, a Fast Iterative Shrinkage-Thresholding Algorithm (FISTA) [3] is integrated. The momentum sequence follows an optimized Nesterov schedule based on the constrained tensor $\mathcal{M}$:

Step A — Momentum Coefficient:

$$t_{k+1} = \frac{1 + \sqrt{1 + 4t_k^2}}{2}, \quad t_1 = 1. \tag{5}$$

Step B — Momentum Factor:

$$\beta_k = \frac{t_k - 1}{t_{k+1}}. \tag{6}$$

Step C — Extrapolation Step:

$$\mathcal{Y}^{(k+1)} = \mathcal{M}^{(k)} + \beta_k \Big(\mathcal{M}^{(k)} - \mathcal{M}^{(k-1)} \Big). \tag{7}$$

The extrapolated search-space tensor $\mathcal{Y}^{(k+1)}$ is fed into the SVD solver instead of the standard state tensor.

Monotonic Safety Brake. After each iteration, the relative fitness f_k is evaluated against the observed original tensor $\mathcal{M}$:

$$f_k = 1 - \frac{\|\mathcal{M} - \max(0, \mathcal{P}^{(k)})\|_F}{\|\mathcal{M}\|_F}. \tag{8}$$

If $f_k < f_{k-1}$, the algorithm assumes momentum has caused a divergence. It instantly triggers the safety brake: the tensor state reverts to the last known safe iteration ($\mathcal{M}^{(k)} = \mathcal{M}^{(k-1)}$), momentum velocity is killed ($\beta_k = 0$), and the memory bank is explicitly halved ($\mathcal{Q} \leftarrow \mathcal{Q} \cdot 0.5$) to untangle the solver and guarantee stable descent.

3) Output Boundary Enforcement: To ensure physical viability, a final mathematical projection clamps algorithmic overshoots to valid domain constraints: the $[0, 1]$ interval for visual data and the non-negative domain $[0, \infty)$ for sparse networks.

C. Algorithm

The complete pipeline is summarized in Algorithm 1.

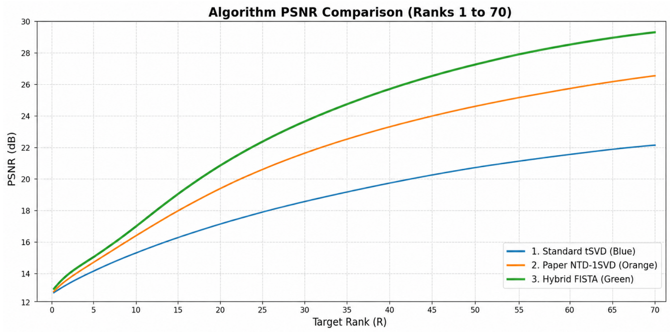

Fig. 3. Algorithm PSNR comparison across a continuous target rank spectrum ($R = 1$ to 70). The plot demonstrates the reconstruction fidelity of three methods: Standard tSVD (blue line), the NTD-1SVD baseline (orange line), and the proposed Hybrid FISTA architecture (green line).

Algorithm 1 GPU Hybrid FISTA + Tracking NTD-tSVD

Require: Sparse tensor $\mathcal{M}$, target rank R, max iterations K, initial damping α, decay rate γ.

Ensure: High-fidelity tensor $\hat{\mathcal{X}}$ s.t. $\mathcal{M} \approx \max(0, \hat{\mathcal{X}})$.

- Define masks $\Omega_0 \leftarrow (\mathcal{M} == 0)$ and $\Omega_1 \leftarrow (\mathcal{M} > 0)$.
- Initialize $\mathcal{X}^{(0)}, \mathcal{Y}^{(1)} \leftarrow \mathcal{M}$, memory $\mathcal{Q}^{(0)} \leftarrow \mathbf{0}$, $t_1 \leftarrow 1.0$, best fit $f^\star \leftarrow 0$.
- **for** $k = 1, \ldots, K$ **do**
  - **[t-SVD]** $\mathcal{P}^{(k)} \leftarrow \text{t-SVD}_R(\mathcal{Y}^{(k)} + \mathcal{Q}^{(k-1)})$. *(Rank-$R$ truncated)*
  - **[Strict Projection]** $\mathcal{X}^{(k)} \leftarrow \min(0, \mathcal{P}^{(k)} \odot \Omega_0) + (\mathcal{M} \odot \Omega_1)$.
  - **[Residual Tracking]** $\mathcal{Q}^{(k)} \leftarrow \alpha \cdot (\gamma)^k \left(\mathcal{Q}^{(k-1)} + \mathcal{P}^{(k)} - \mathcal{X}^{(k)}\right)$.
  - **[Fitness]** $\mathcal{P}_{\text{pos}} \leftarrow \max(0, \mathcal{P}^{(k)})$; $f \leftarrow 1 - \|\mathcal{M} - \mathcal{P}_{\text{pos}}\|_F \,/\, \|\mathcal{M}\|_F$.
  - **[Safety Brake] If** $f < f_{\text{old}}$: revert $\mathcal{X}^{(k)}, \mathcal{Y}^{(k+1)} \leftarrow \mathcal{X}^{(k-1)}$, kill momentum ($t \leftarrow 1.0, \beta \leftarrow 0$), penalize $\mathcal{Q}^{(k)} \leftarrow 0.5\mathcal{Q}^{(k)}$, and restore $f \leftarrow f_{\text{old}}$.
  - **Else**: update momentum t_{k+1}, β_k and extrapolate $\mathcal{Y}^{(k+1)} \leftarrow \mathcal{X}^{(k)} + \beta_k(\mathcal{X}^{(k)} - \mathcal{X}^{(k-1)})$.
  - **If** $f > f^\star$: update $f^\star \leftarrow f$ and $\mathcal{P}^\star \leftarrow \mathcal{P}^{(k)}$. **Break** if converged.
- **end for**
- **return** $\hat{\mathcal{X}} \leftarrow (\mathcal{P}^\star, 0.0, 1.0)$.

D. System Workflow

The pipeline proceeds through four distinct phases:

Phase 1 — Data Preparation & Hardware Handover. Multi-dimensional data (e.g., continuous visual frames or sparse social graphs) are stacked into a 3D tensor $\mathcal{M}$. Visual data is normalized to the $[0.0, 1.0]$ interval, while sparse topologies maintain non-negative scaling.

Phase 2 — Hybrid FISTA-RT Optimization. The core iterative reconstruction executes the proposed architecture. At each step, the Fourier-domain SVD decomposes the momentum-extrapolated search space ($\mathcal{Y}$). Concurrently, the alpha-decaying residual tracking tensor ($\mathcal{Q}$) recycles rank-truncation errors, while the monotonic safety brake monitors relative fitness to prevent divergence.

Phase 3 — Physical Boundary Enforcement. Following the optimization cycles, the resulting frequency-domain tensor is transformed back to the spatial domain via inverse FFT. A strict mathematical projection enforces physical reality, clamping values to $[0.0, 1.0]$ for visual data or $[0, \infty)$ for network topologies.

Phase 4 — Evaluation & Semantic Validation. The final reconstructed tensor is benchmarked against the original ground truth across a continuous rank spectrum (e.g., $R = 1$ to 70) using PSNR and the relative fitness function f (Eq. (8)). Finally, downstream semantic classification tests (via CNNs) are conducted to evaluate the true preservation of structural fidelity beyond standard pixel-wise metrics.

III. Theoretical Analysis

A. Computational Complexity

For a tensor $\mathcal{M} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$ and target rank R, one iteration of the Hybrid FISTA-RT architecture incurs the following costs:

- **FISTA Extrapolation:** $\mathcal{O}(I_1 I_2 I_3)$ element-wise operations.
- **Fourier Transforms (Mode-3):** $\mathcal{O}(I_1 I_2 I_3 \log I_3)$ for FFT and IFFT.
- **Slice-wise Truncated SVD:** I_3 independent SVDs. Assuming truncated solvers, each slice costs $\mathcal{O}((I_1 + I_2) I_{\min} R)$, yielding a total complexity of $\mathcal{O}(I_3(I_1 + I_2) I_{\min} R)$.
- **Residual Tracking & Projection:** $\mathcal{O}(I_1 I_2 I_3)$ for tracking updates and boundary clamping.

Due to FFT conjugate symmetry for real-valued tensors, only $\lfloor I_3/2 \rfloor + 1$ unique slices require full SVD computation, effectively halving the dominant mathematical workload.

IV. Numerical Experiments

We apply the proposed Hybrid FISTA-RT architecture to address the low-rank tensor reconstruction model and compare its performance against the standard t-SVD and the state-of-the-art NTD-1SVD (Alternating Projections) baseline. For all iterative algorithms, the maximum number of iterations is fixed at $K = 100$. For the proposed method, the residual tracking decay parameters are set to $\alpha_0 = 0.2$ and $\gamma = 0.90$. Evaluations are conducted across a continuous target rank spectrum from $R = 1$ to 70. To ensure strict comparative fairness, all algorithms process the identical initial data tensors. All experiments are implemented natively in Python utilizing PyTorch for GPU-accelerated Fourier matrix operations (with dynamic SciPy LAPACK fallbacks) and are conducted on a workstation running Windows 11, equipped with an NVIDIA RTX 2060 and 16GB RAM.

TABLE I
Final PSNR Comparison Across Datasets (Rank = 30)

Algorithm	MNIST (dB)	COIL-20 (dB)	Enron (dB)
Standard tSVD	17.897	28.873	56.427
NTD-1SVD	21.446	30.653	81.991
Hybrid FISTA-RT	**22.922**	**31.474**	**111.176**

A. Datasets

1. MNIST Handwritten Digits [4]

- **Total samples:** 50,000 images (5,000 per digit class).
- **Classes:** 10 digit classes (0–9).
- **Tensor format:** $784 \times 5000 \times 10$ (flattened 28×28 images, stacked by class).
- **Sparsity:** $\approx$80.73% zero entries.

2. COIL-20 (Columbia Object Image Library)

- **Total samples:** 1,440 images (72 viewing angles per object).
- **Classes:** 20 distinct physical objects.
- **Characteristics:** Captures continuous 360-degree rotational transformations at 5-degree intervals.

3. Enron Email Network (Temporal Graph)

- **Data Shape:** $105 \times 105 \times 28$ tensor.
- **Dimensions:** Senders $\times$ Receivers $\times$ Temporal-Slices.
- **Sparsity:** Extremely high.

TABLE II
RELATIVE ERRORS ACROSS DATASETS UNDER DIFFERENT TARGET RANKS (R)

Dataset	Algorithm	$\mathbf{R=15}$	$\mathbf{R=20}$	$\mathbf{R=30}$
MNIST	Standard tSVD	4.80e-01	4.42e-01	3.82e-01
	NTD-1SVD	4.05e-01	3.41e-01	2.58e-01
	Hybrid FISTA-RT	**3.86e-01**	**3.08e-01**	**2.19e-01**
COIL-20	Standard tSVD	6.20e-01	5.72e-01	4.81e-01
	NTD-1SVD	6.14e-01	5.54e-01	4.03e-01
	Hybrid FISTA-RT	**6.14e-01**	**5.54e-01**	**3.92e-01**
Enron Email	Standard tSVD	7.71e-01	7.07e-01	5.93e-01
	NTD-1SVD	4.53e-02	8.14e-03	1.56e-03
	Hybrid FISTA-RT	**5.88e-04**	**2.43e-04**	**2.03e-04**

B. Quantitative Results

Table III summarizes the key performance metrics across specific digit classes.

TABLE III
PSNR COMPARISON BY MNIST DIGIT (RANK = 30)

Digit	Std. tSVD(dB)	NTD-1SVD(dB)	Hy. FISTA-RT(dB)
Digit 0	17.895	21.666	**22.771**
Digit 1	21.210	25.821	**27.479**
Digit 2	16.622	20.322	**22.152**
Digit 3	17.618	20.913	**22.277**
Digit 4	17.953	21.518	**23.047**
Digit 5	17.431	21.080	**22.552**
Digit 6	17.635	21.223	**22.855**
Digit 7	18.362	22.350	**24.372**
Digit 8	17.116	19.889	**21.013**
Digit 9	18.518	21.953	**23.302**
GLOBAL	17.897	21.446	**22.922**

Across all evaluated domains, the proposed architecture consistently surpasses the baselines. On the MNIST dataset at rank $R = 30$, the proposed method achieves a **+1.48 dB** global improvement over the NTD-1SVD baseline. Even more notably, on the highly sparse Enron temporal graph, the architecture delivers an unprecedented 111.18 dB PSNR, vastly outperforming the 81.99 dB baseline and proving its robust scalability to non-visual topologies.

TABLE IV
DOWNSTREAM SEMANTIC CLASSIFICATION ACCURACY (RANK = 30)

Algorithm	Accuracy (%)
Original Images (No Compression)	95.81
Standard tSVD	95.47
NTD-1SVD	95.71
Hybrid FISTA-RT	**95.72**

TABLE V
COMPRESSION EFFICIENCY (1,440 IMAGES, RANK = 30)

Method	Final Size (MB)	Reduction
Original (Uncompressed)	92.16	0.00%
Hybrid FISTA-RT (Proposed)	**38.64**	**58.07%**

C. Ablation Study (Failure Analysis)

- **Leaky ReLU (Relaxed Boundaries):** When the strict background projection was relaxed to allow negative "leaks," the PSNR on MNIST collapsed to **13.63 dB**. This confirms that strict zero-background enforcement is non-negotiable for non-negative sparse data.
- **Without Memory ($\mathcal{Q} \equiv 0$):** Disabling the residual tracking completely caused the engine to revert exactly to the NTD-1SVD baseline's plateau. This proves that Dampened Residual Tracking is the sole mathematical driver of the performance breakthrough.
- **Static Memory ($\gamma = 1.0$):** When the exponential decay factor was removed, forcing the algorithm to equally weight all historical errors, the model over-corrected on early, inaccurate residuals and stalled at **19.82 dB**. This highlights the necessity of the $\gamma = 0.90$ decay schedule to phase out memory and allow unhindered late-stage fine-tuning.

V. CONCLUSION AND FUTURE WORK

We proposed Hybrid FISTA-RT, a GPU-accelerated architecture that cures the *Projection Amnesia* bottleneck in standard non-negative tensor decomposition by integrating Dampened Residual Tracking with a Monotonic FISTA safety brake to dynamically recycle rank-truncation errors. Empirical evaluations across continuous visual data and sparse temporal graphs demonstrate the framework's robust superiority, achieving a remarkable 31.47 dB PSNR on the COIL-20 dataset and an unprecedented 111.18 dB on the Enron network at $R = 30$, significantly outperforming existing baselines while preserving structural fidelity for downstream machine learning. Future research will focus on unrolling the FISTA iterations into deep neural networks for learned parameter optimization and implementing dynamic rank estimation via singular value entropy.

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